

Weather Data Analysis Report

1. Project Title

Weather Data Analysis Using Python

2. Objective

The objective of this project is to analyze weather data using Python and identify trends in temperature, humidity, and rainfall. The project also aims to visualize the data using graphs for better understanding of seasonal weather patterns.

3. Tools and Technologies Used

- Python
- Google Colab
- Pandas
- Matplotlib
- Seaborn

4. Dataset

- Dataset Name: **climate_data.csv**
- The dataset contains weather information such as:
 - Date
 - Average Temperature
 - Humidity
 - Rainfall
 - Wind Speed
 - Atmospheric Pressure

5. Data Preprocessing

The following preprocessing steps were performed:

- Loaded the dataset using Pandas.
- Checked for missing values.
- Removed duplicate records.
- Converted the Date column into datetime format.
- Verified that the dataset was clean and ready for analysis.

6. Data Analysis

The following analyses were performed:

- Temperature trend analysis
- Humidity trend analysis
- Monthly rainfall analysis
- Monthly average temperature comparison

7. Visualizations

The following charts were created:

- Temperature Trend Graph
- Humidity Trend Graph
- Monthly Average Temperature Bar Chart
- Monthly Rainfall Bar Chart